

Muhammad Patel

Aerospace Engineering Project Portfolio

BEng Aerospace Engineering with Pilot Studies | University of Central Lancashire | Graduated 2025

A broad set of projects covering CAD, mechanical design, aircraft design, CFD, FEA, motion analysis, manufacturing, testing, automation, project management, and technical communication.

SolidWorks

Ansys Fluent

FEA

Motion

3D Printing

VBA API

13

project snapshots included

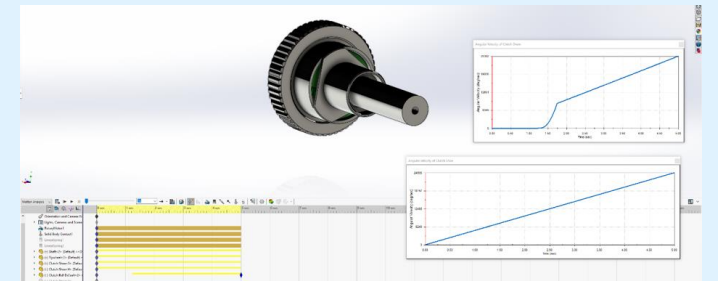
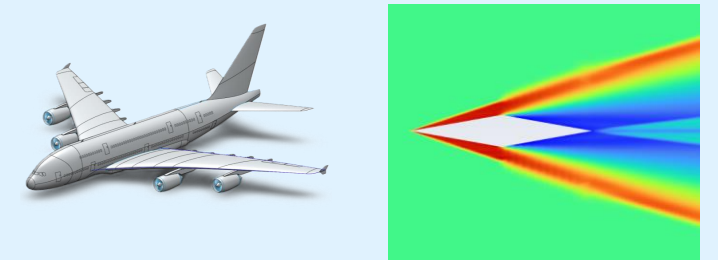
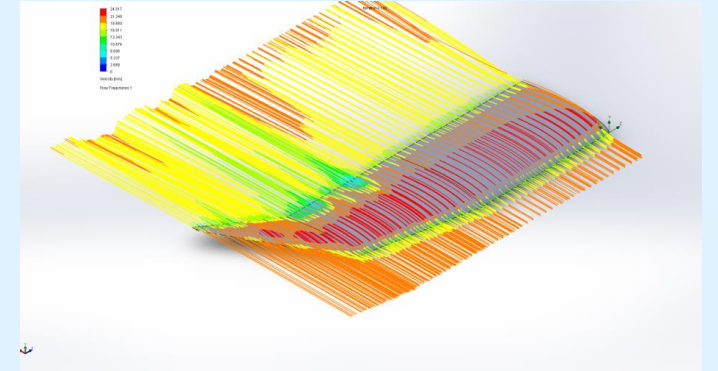
CAD + tests

from models to validation

Analysis led

simulation, data and decisions

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Projects now included

A wider portfolio showing design, analysis, manufacture, testing, automation and project management evidence.

01	Winglet wake-vortex dissertation	CAD, CFD, 3D printing, wind tunnel validation	02	Clutch design and motion analysis	CAD assembly, drawings, FEA, motion study
03	SolidWorks API gear generator	VBA, API automation, validation, Excel presets	04	Aircraft design and stability	Payload-range, constraints, sizing, stability
05	Turbine blade aeromechanics	FEA, modal, fatigue, material comparison	06	A380 surface modelling	Advanced surfacing, NURBS, continuity
07	Supersonic aerofoil CFD	Mach, AoA, shockwaves, roughness, ANSYS Fluent	08	Aircraft wing design and manufacture	Spars, ribs, testing, manufacturability
09	Structures and FEA wing study	Torsion, bending, aircraft wing FEA, validation	10	Engineering science lab portfolio	Fabrication, Reynolds, Bernoulli, tensile/composite
11	Project management analysis	Stakeholders, WBS, OBS, earned value, risk	12	Challenger risk case study	Communication, safety culture, governance



Winglet Design and Wake Vortex Analysis

Aerospace research project combining CAD, CFD, 3D printing and wind tunnel validation.

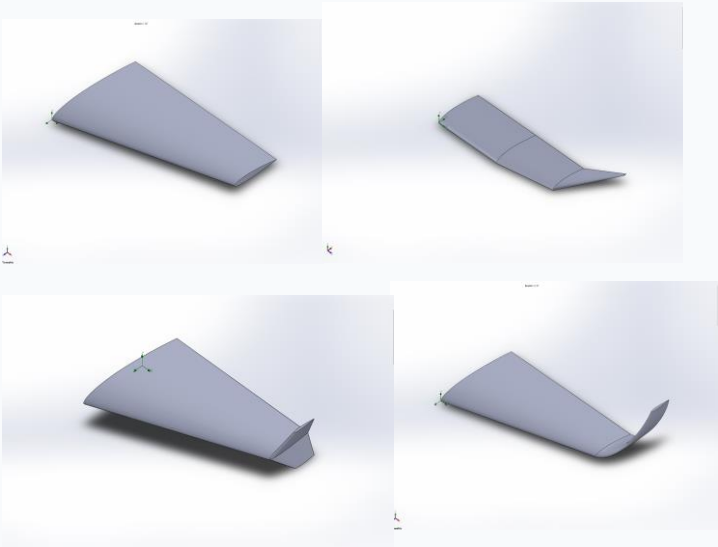
- SolidWorks
- Ansys Fluent
- CFD
- Wind tunnel
- 3D printing

What I did

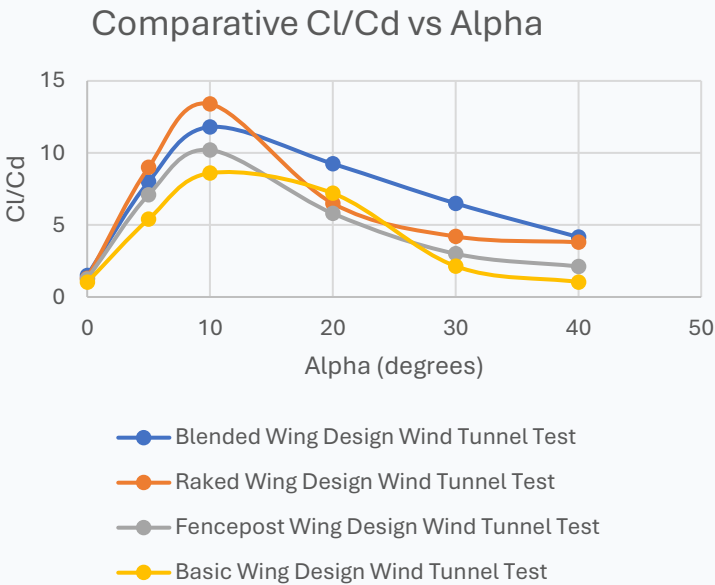
- Compared blended, raked and fencepost wingtips against a flat baseline.
- Ran CFD and validated trends using 3D-printed wind tunnel models.
- Used vorticity, lift, drag and CL/CD to compare aerodynamic performance.

Result and judgement

- Raked wingtip gave the best aerodynamic efficiency.
- Blended winglet produced the weakest peak vortex.
- Project showed the value of combining simulation with physical testing.



Wingtip concepts compared under the same conditions.



Dissertation evidence combining CAD, CFD and practical testing.

4

wingtip designs

20%

peak vorticity reduction

13.4

maximum CL/CD

0.4 mm

selected CFD mesh

Clutch Design and Motion Analysis

SolidWorks mechanical design project for a centrifugal clutch assembly.

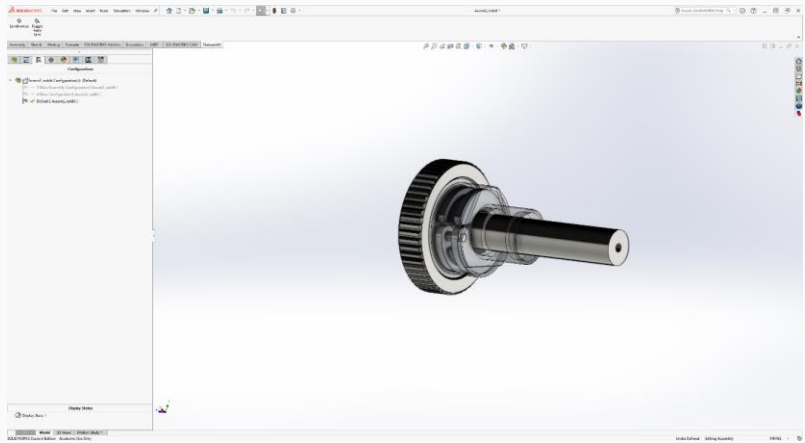
- SolidWorks
- Motion analysis
- FEA
- Drawings
- Honda GX

Project scope

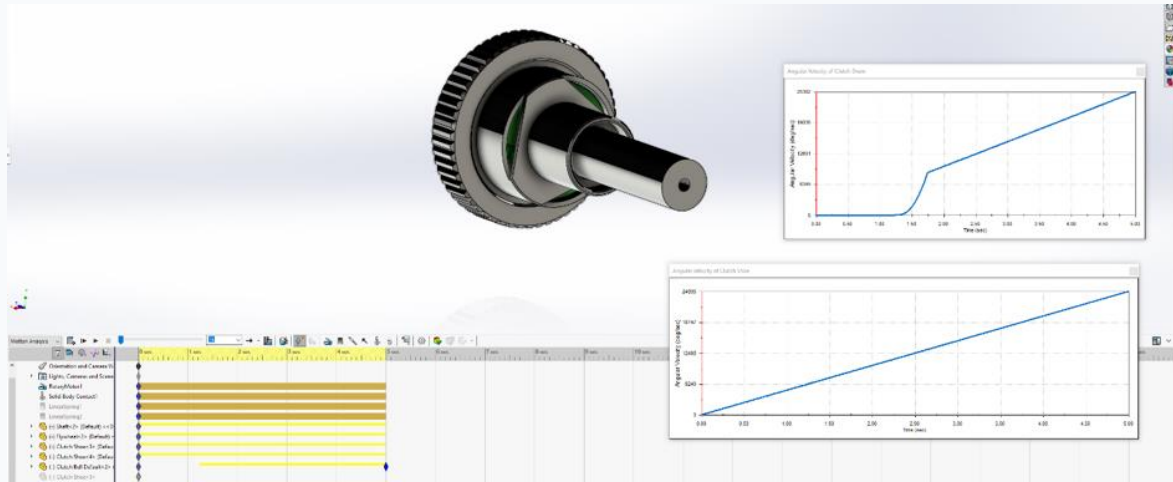
- Modelled shaft, flywheel, clutch bell and clutch shoe.
- Compared 2, 3 and 4 shoe assembly configurations.
- Produced 2D engineering drawings and assembly documentation.

Analysis work

- FEA used to assess stress, strain and displacement on the clutch shoe.
- Motion analysis used to study engagement behaviour as RPM increased.
- Realistic Honda GX120 values used for torque and power context.



SolidWorks component models and assembly evidence.



Motion analysis output showing engagement response.

4

main components

2,3,4

shoe layouts

7.3 Nm

GX120 torque reference

2.4 kW

GX120 power reference

SolidWorks API Gear Generator

A VBA and SolidWorks API tool for automatic parametric spur gear generation.

VBA	SolidWorks API	Excel presets	Validation
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Automation workflow

- Generated spur gears from custom user inputs or Excel presets.
- Built base, teeth, circular pattern, chamfers, hub and bore features.
- Used a modular structure so the same build logic worked for both modes.

Engineering value

- Reduced repetitive manual modelling time by over 90%.
- Added input checks for tooth count, bore, hub and dimensions.
- Demonstrated programming as a practical CAD productivity tool.

```
'''
inputStr = InputBox("Enter number of teeth (Z):", "Gear Setup", "")
If inputStr = "" Then Exit Function
If IsNumeric(inputStr) Then
    If CInt(inputStr) = Int(CInt(inputStr)) And CInt(inputStr) >= 0 Then
        Z = CInt(inputStr)
        Exit Do
    End If
End If
MsgBox "Invalid input. Enter a whole number greater than 0."
Loop

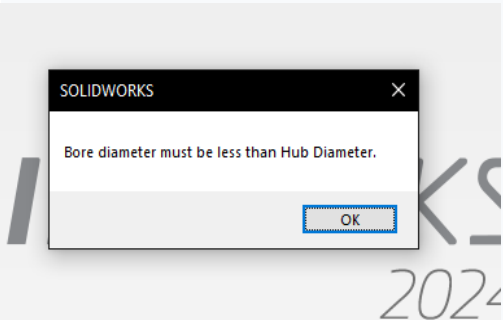
Do
    inputStr = InputBox("Enter module size in mm:", "Gear Setup", "")
    If inputStr = "" Then Exit Function
    If IsNumeric(inputStr) Then
        If Val(inputStr) > 0 Then
            m_m = Val(inputStr)
            Exit Do
        Else
            MsgBox "Invalid module size. Enter a number greater than 0."
        End If
    End If
Loop
ModuleSize = m_m / 1000

Do
    inputStr = InputBox("Enter hub diameter d (mm):", "Gear Setup", "")
    If inputStr = "" Then Exit Function
    If IsNumeric(inputStr) Then
        hubDia = Val(inputStr) / 1000
        Exit Do
    End If
End If
MsgBox "Invalid hub diameter."
Loop

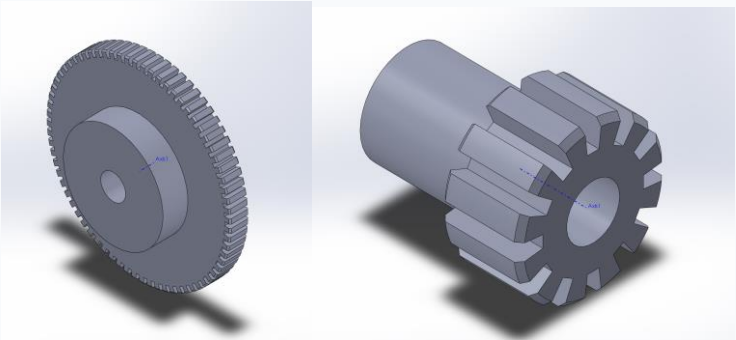
Do
    inputStr = InputBox("Enter Bore diameter D (mm):", "Gear Setup", "")
    If inputStr = "" Then Exit Function
    If IsNumeric(inputStr) And Val(inputStr) < hubDia * 1000 Then
        boreDia = Val(inputStr) / 1000
        Exit Do
    Else
        MsgBox "Bore diameter must be less than Hub Diameter."
    End If
End If
Loop

Do
    inputStr = InputBox("Enter shoulder height h (mm):", "Gear Setup", "")
    If inputStr = "" Then Exit Function
    If IsNumeric(inputStr) And Val(inputStr) > 0 Then
        hubShoulder = Val(inputStr) / 1000
        Exit Do
    Else
        MsgBox "Invalid hub thickness."
    End If
End If
Loop

Do
    inputStr = InputBox("Enter total hub thickness H (mm):", "Gear Setup", "")
    If inputStr = "" Then Exit Function
    If IsNumeric(inputStr) And Val(inputStr) > hubShoulder * 1000 Then
        hubTotal = Val(inputStr) / 1000
        Exit Do
    Else
        MsgBox "Hub Thickness should be greater than shoulder height."
    End If
End If
Loop
```



Validation logic and user warning example.



Generated custom and preset gear outputs.

90%+

time reduction

2

input modes

3 to 100

tooth-count range

0

validated rebuild errors

Aircraft Design and Stability Mission Analysis

Conceptual civil transport aircraft sizing and stability study.

Payload-range

Constraint analysis

Stability

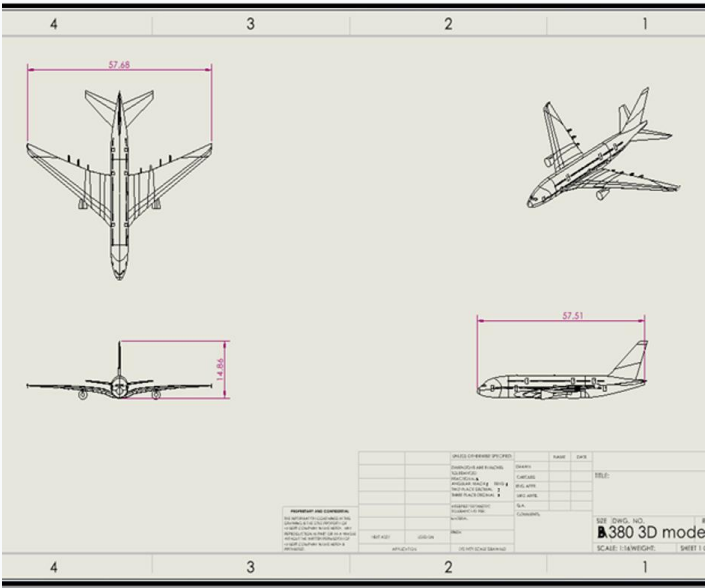
Excel

Key work

- Analysed a Southampton to Lyon-Bron civil transport mission.
- Used payload-range and constraint analysis to select feasible sizing values.
- Calculated MAC, CG, aerodynamic centre, neutral point and landing gear references.

Design judgement

- Sanity-checked early spreadsheet outputs against real aircraft benchmarks.
- Used A330-class engine and wide-body scale values for a more realistic concept.
- Individual task adjusted seating layout and weight distribution.



Aerodynamic Stability Shown on Plane

265

baseline passengers

~230t

realistic MTOW scale

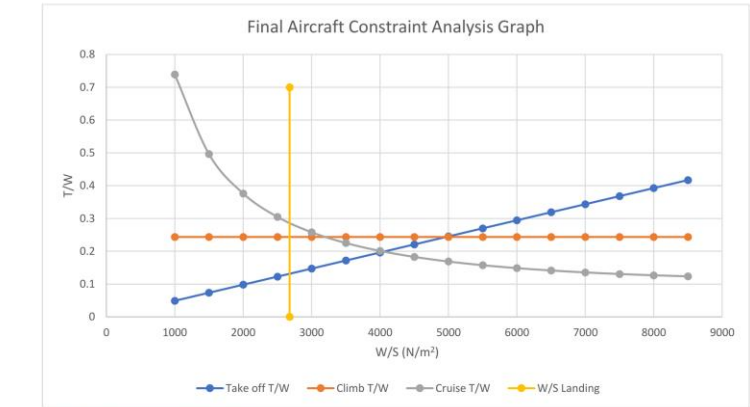


Figure 28: Final Aircraft Constraint Analysis

Constraint analysis used to define feasible design points.

~6200

N/m² wing loading

7.38 m

MAC reference

Turbine Blade Aeromechanics Analysis

Simulation-led study of turbine blade materials, modal response, fatigue and optimisation.

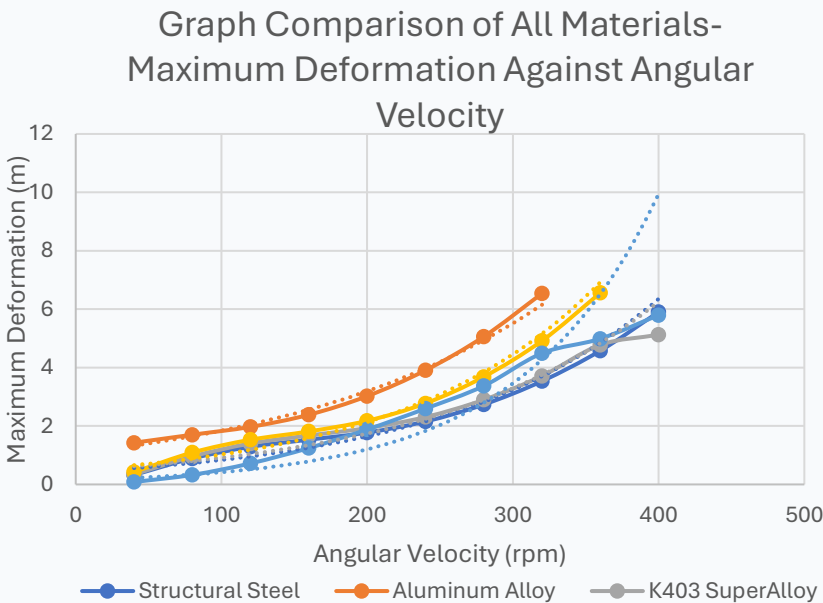
- FEA
- Mesh study
- Modal
- Fatigue
- Optimisation

Simulation workflow

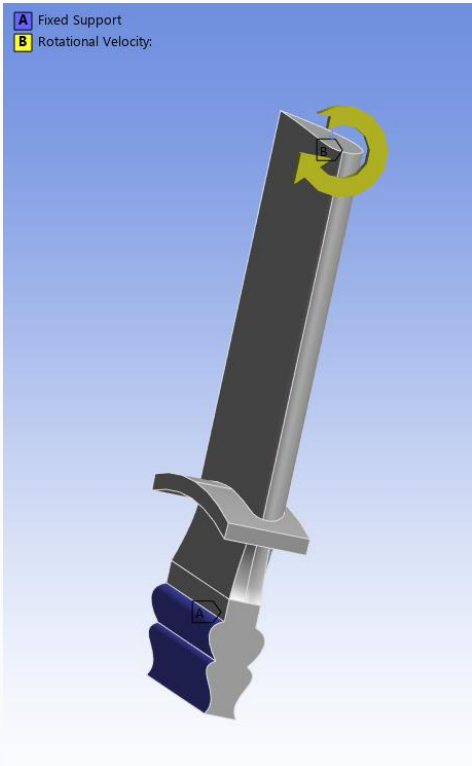
- Performed mesh verification to balance accuracy and solver time.
- Compared structural steel, aluminium, K403, titanium and graphene behaviours.
- Ran modal and high-cycle fatigue studies on the selected material.

Key findings

- K403 superalloy selected for later modal and fatigue studies.
- Mode 23 produced the highest modal deformation at 41.948 Hz.
- Hollow blade and fillets produced the largest stress reductions.



Material comparison across stress, strain and deformation outputs.



Boundary Conditions for High-Cycle Fatigue Testing

0.4 m

selected mesh size

K403

selected superalloy

42.96%

stress reduction

41.948 Hz

critical mode

A380 Surface Modelling

SolidWorks surface modelling project focused on complex aircraft-style geometry.

Surface modelling

Lofted surface

Boundary surface

NURBS

Tools applied

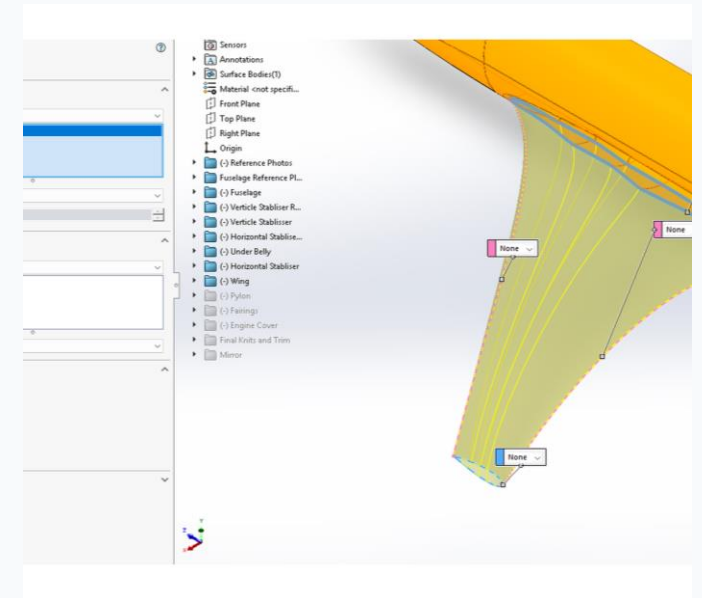
- Used lofted, boundary, filled, trimmed, knit and split-line surface workflows.
- Reviewed NURBS and G0, G1, G2 continuity concepts.
- Refined complex aircraft geometry into a cleaner surface model.

Skills shown

- Advanced CAD modelling beyond standard solid features.
- Understanding of smooth transitions and aerodynamic-style surfaces.
- Comparison of SolidWorks against CATIA, Rhino, Fusion and AutoCAD.



Completed A380 surface model.



Boundary surface workflow on aircraft geometry.

6

surfacing tools

G0/G1/G2

continuity reviewed

NURBS

surface theory

Aerospace

freeform geometry

Supersonic Aerofoil CFD Investigation

Aeromechanics CFD project studying shockwaves, pressure distribution and geometry effects.

ANSYS Fluent

Mach flow

Shockwaves

AoA

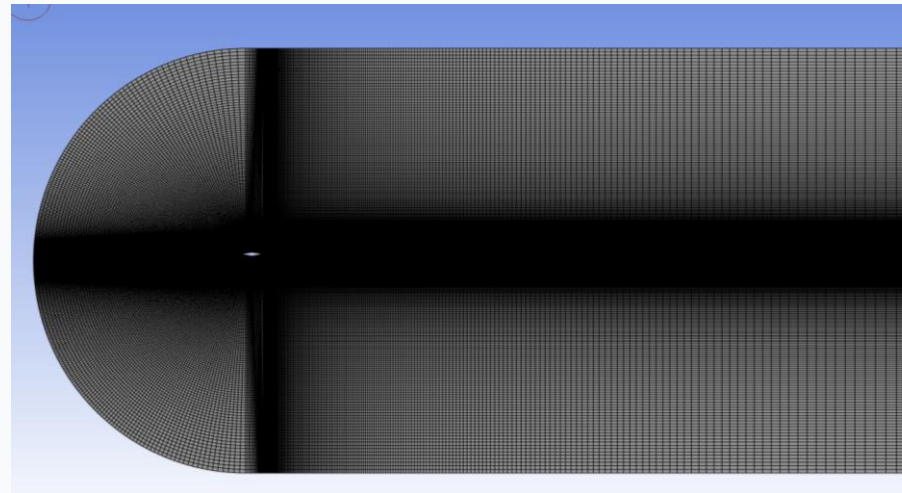
Roughness

What I investigated

- Tested double-wedge, bi-convex and dome-shaped aerofoils.
- Compared Mach number, angle of attack, roughness and geometry variations.
- Used shadowgraph images and pressure taps for the wind tunnel lab section.

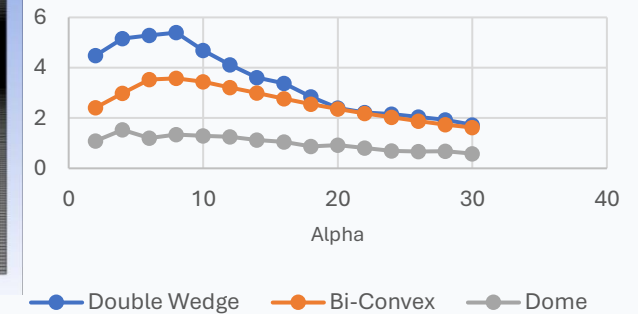
Engineering learning

- At 5° AoA, asymmetric shock and pressure patterns created lift.
- Bi-convex profile showed the strongest overall CL/CD trend across most cases.
- Small geometry changes had large effects in supersonic flow.



Mesh Refinement Image of Optimum Mesh

Graph of CL/CD Against Angle of Attack- Comparative



Graph of CL/CD Against Angle of Attack- Comparative

Mach 1.4

wind tunnel test

Mach 1 to 6

CFD speed range

3

main aerofoil shapes

7

test categories

Aircraft Wing Design and Manufacture

Group project designing, building and testing a lightweight aircraft wing.

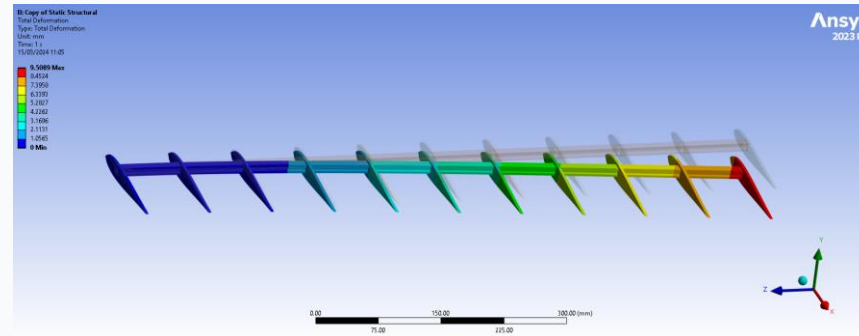
[Spar design](#)[Ribs](#)[Manufacturing](#)[Testing](#)[DFM](#)

Design brief

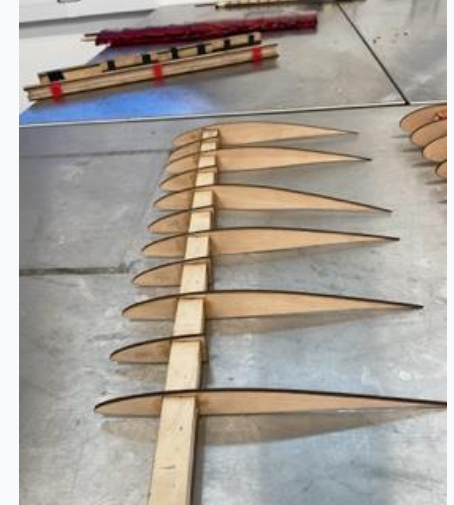
- Designed a wing to support a 3 kg fuselage and survive repeat flight attempts.
- Balanced glide distance, 180° turn performance, transportability and manufacturability.
- Developed spar, rib, stringer, skin and aileron design iterations.

Test and iteration

- Square spar, I-beam spar and slotted I-beam concepts were compared.
- I-beam solution achieved performance with a 42% weight reduction against the square spar.
- Manufacturing constraints drove practical redesign decisions.



I-beam spar testing and deflection comparison.



Wing assembly with spar and ribs.

3 kg

fuselage requirement

9 kg

minimum spar support

42%

I-beam mass reduction

4

major design iterations

Subsonic Aircraft Wing FEA and Lab Testing

Coursework linking torsion, bending experiments and FEA analysis of an aircraft wing model.

Torsion

Bending

FEA

Material comparison

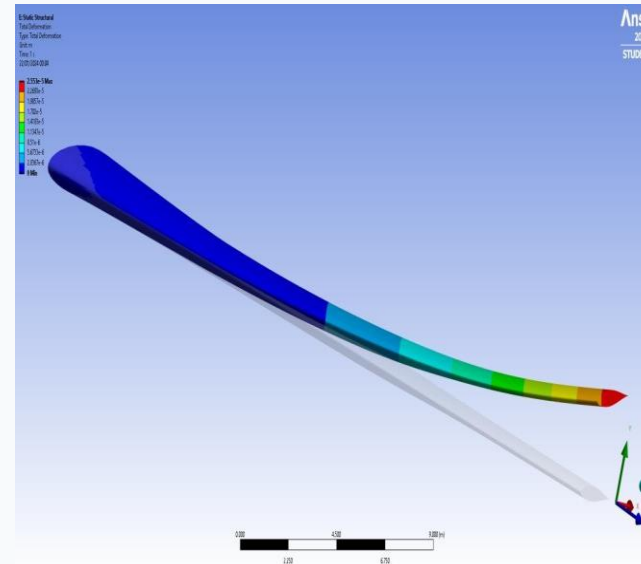
Validation

Lab and analysis work

- Tested solid and hollow brass sections under torsion.
- Measured beam deflection across several materials and dimensions.
- Analysed a subsonic aircraft wing model under force, moment and velocity loading.

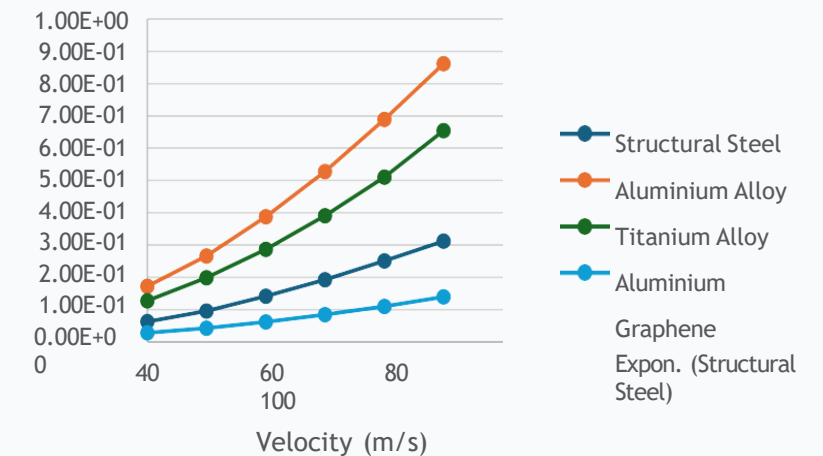
Engineering judgement

- Linked test trends to theory such as Hooke's Law and cantilever beam behaviour.
- Compared material response under dynamic pressure conditions.
- Reflected on FEA limitations when geometry and assumptions affect credibility.



Wing geometry, fixed support and deformation outputs.

Graph of Total Maximum Deformation Against Velocity



FEA force, moment and material trend graphs.

FEA

model setup

torsion

lab testing

bending

material response

validation

simulation limits

Stakeholder, Scheduling and Earned Value Analysis

Project management coursework covering governance, planning, scheduling, risk and earned value.

Stakeholders

WBS

OBS

Risk

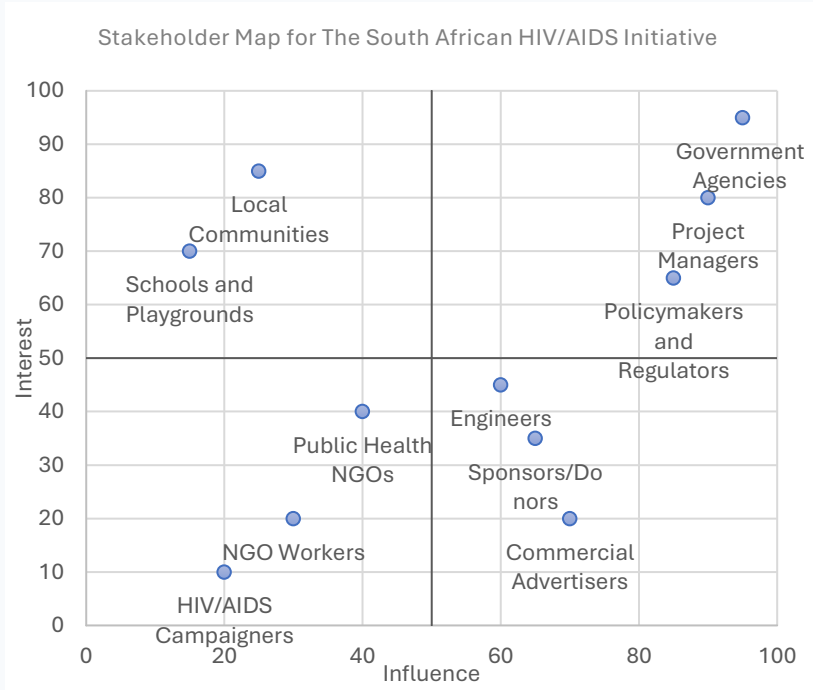
Earned value

Analysis scope

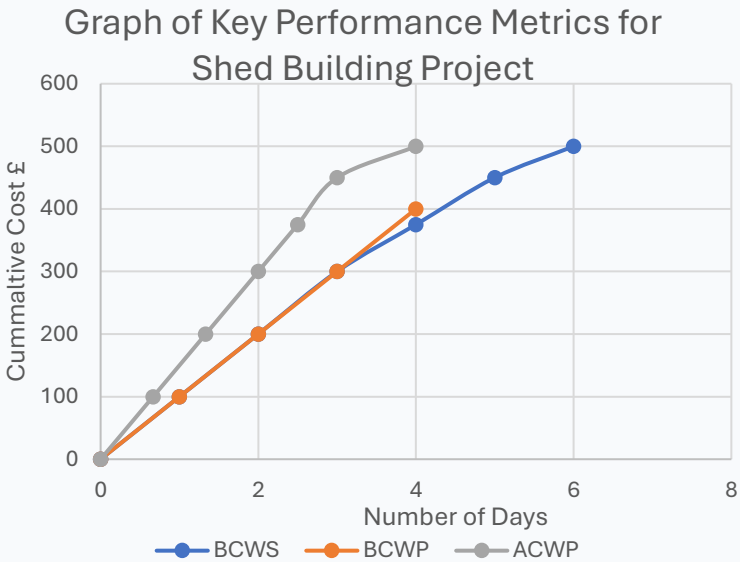
- Stakeholder analysis for a South African HIV/AIDS roundabout initiative.
- Planning and scheduling exercise with assumptions, duration estimates and networks.
- Earned value analysis applied to a shed-building project.

Skills shown

- Structured communication of stakeholder needs and governance roles.
- Use of WBS, OBS and integrated responsibility planning.
- Data-led understanding of schedule and cost performance.



Influence and interest stakeholder map.



Earned value performance metrics.

WBS

work breakdown

OBS

organisation structure

EVA

cost and schedule metrics

risk

mitigation planning

Challenger Disaster Risk and Communication Analysis

Project management case study examining whether Challenger was a technical failure or management failure.

Risk

Communication

Safety culture

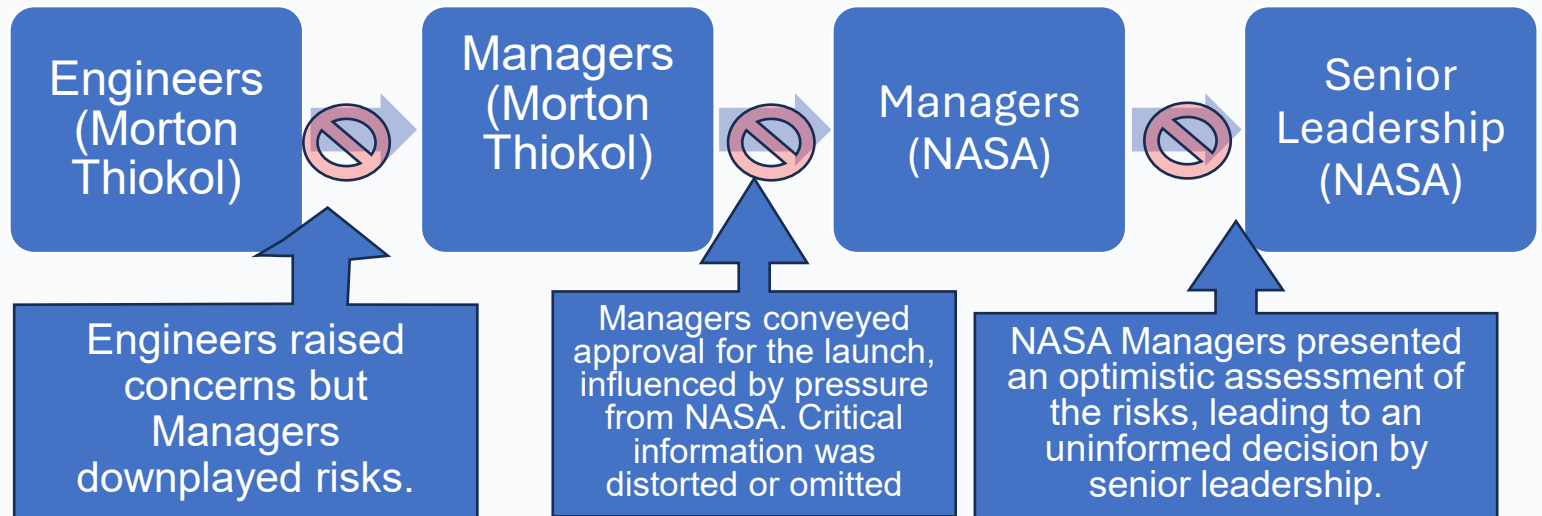
Governance

Case focus

- Examined O-ring failure in the context of organisational decision-making.
- Explored communication breakdowns between engineers, managers and leadership.
- Assessed stakeholder pressure, risk escalation and independent oversight.

Learning value

- Technical risks need clear escalation routes and independent review.
- Past success should not normalise deviations from safe performance.
- Engineering decisions need both technical evidence and strong governance.

**73 s**

failure after lift-off

risk

escalation focus

safety

culture analysis

PM

management lens

Engineering Science Lab Portfolio

First-year laboratory portfolio covering fabrication, fluids, materials and composite testing.

Fabrication

Reynolds

Bernoulli

Tensile

Composites

Lab areas

- Fabrication workshop using bending and CNC-related setup evidence.
- Fluid mechanics experiments including Reynolds and Bernoulli principles.
- Tensile and composite testing used to analyse material behaviour.

Skills developed

- Measurement, data handling, graphing and technical reporting.
- Connecting theory to practical laboratory observations.
- Early foundation for later CAD, FEA and simulation projects.



Figure 10: Example of tool used for bending.



Figure 11: Picture of final product.



Figure 12: Picture of tool on machine with backmarkers shown.

Fabrication workshop evidence.

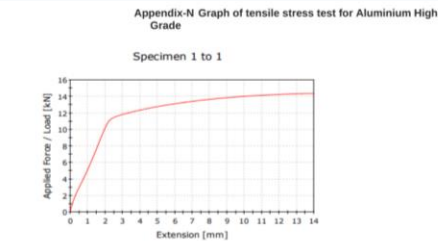


Figure 4 Graph of tensile stress test for Aluminium High Grade

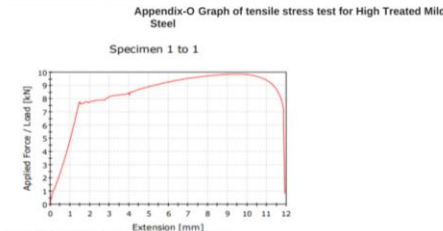


Figure 5 Graph of tensile stress test for High Treated Mild Steel

Appendix-P Tensile Sample Results for ER1030
Table 4: Tensile Sample Results for ER1030

Material	Pre-test Width (mm)	Pre-test Thickness (mm)	Width After Test (mm)	Thickness After Test (mm)
Mild Steel	13.0	2.0		
Mild Steel (Treated)	13.0	2.0	11.0	1.5
Aluminium 1050 Grade	13.0	2.0		

Material testing graphs and results evidence.

5

lab reports

fluids

Reynolds and Bernoulli

materials

tensile/composite

workshop

fabrication skills

What the full project set demonstrates

Grouped evidence across design, analysis, practical testing and delivery.

Design and CAD

- A380 surface modelling
- Clutch assembly
- Aircraft wing design
- Aircraft sizing and stability

Simulation and data

- Winglet CFD
- Supersonic aerofoil CFD
- Turbine FEA
- Aircraft wing FEA

Testing and validation

- Wind tunnel testing
- 3D printed wing models
- Lab torsion and bending
- Prototype manufacturing

Delivery and communication

- Project management
- Risk case studies
- Technical reports
- Portfolio and web presentation

Overall message: the portfolio shows a practical engineering workflow, starting from concept and requirements, through CAD and analysis, then into manufacture, testing, validation and communication.



[Online portfolio](https://mgaliya.github.io/MuhammadPortfolio/)

mgaliya.github.io/MuhammadPortfolio/

Portfolio, CV and project evidence

Project pages, downloadable portfolio and supporting evidence are available online.

I am interested in early-career engineering roles across mechanical design, CAD, simulation, aerospace analysis, manufacturing support, testing and product development.

Email: MGaliya16@gmail.com
LinkedIn: [linkedin.com/in/muhammad-galiya](https://www.linkedin.com/in/muhammad-galiya)
Website: mgaliya.github.io/MuhammadPortfolio/



Scan to view the online portfolio

Thank you